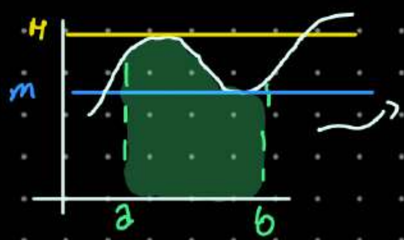
INTroduzione All'integrale

① PROblema Geometrico

Calcolare l'area sotto il grafico di una funzione

per oggi continua e $f(x) > 0$

Su $[a, b]$



$$\int_a^b f(x)\, dx$$

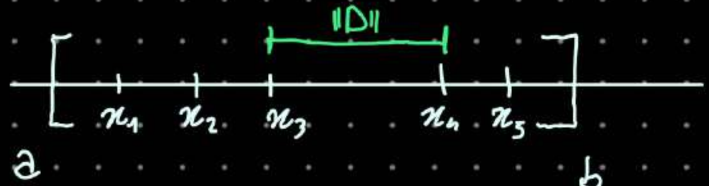
Somma di Riemann → IDea principale

La $f : [a,b] \to \mathbb{R}$ cont

$$m := \inf_{x \in [a,b]} f(x) \quad ; \quad M := \sup_{x \in [a,b]} f(x)$$

- $0 \leq$ Area di f sotto $[a,b]$ $\leq M \cdot (b-a)$
- Dividiamo $[a,b]$ in "sott'intervalli"

$$a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$$



$$D = \{x_0, x_1, \dots, x_n\}$$

$\# D = n$

cardinalità

L' Ampiezza massima della partizione

$$\| D \| := \max_{i=1,\dots,n} \{ \underbrace{x_i, \dots x_{i-1}}_{>0} \}$$

La Somma Inferiore è costruita considerando il minimo (inferiore) della funz. "f" di ciascun SOTTOINTERVALLO della Partizione "D"

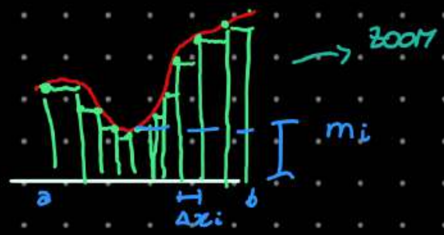
$$m_i := \inf \{ f(x) : x \in [x_{i-1}, x_i] \}$$

Per ogni $i = 1, \dots, n$

La SOMMA INFERIORE rispetto a D è

$$s_i(f, D) := \sum_{i=1}^{n} m_i \Delta x_i$$

dove $\Delta x_i = x_i - x_{i-1}$



$$"\int_a^b f" \geq s(f, D)$$

Definiamo

$$M_i := \sup \{ f(x) : x \in [x_{i-1}, x_i] \}$$

la SOMMA SUPERIORE rispetto a D è

$$S_s(f, D) := \sum_{i=1}^{n} M_i \Delta x_i$$

$$0 \leq "\int_a^b f(x)" \leq M(b-a)$$

le partizioni dell' ∞ [a,b] so Parzialmente ordinate, questo significa che, date due partizioni

P e Q di [a,b]

Possiamo dire che

$$P \leq Q$$

$$P \subseteq Q$$

Se

- $P_1, P_2 \to P_1, P_2 \leq P_1 \cup P_2$
- Se $P \leq Q$

$$s_i(f, P) \leq s_i(f, Q)$$

$\forall \; P, Q$

$$s_i(f, P), \; s_i(f, Q) \leq s_i(f, P \cup Q)$$

- Ha senso definire

$$\underset{\substack{\text{somma}\\ \text{sup}}}{S_s}(f, [a,b]) : \sup_{D} \underset{\text{s inf}}{s_i}(f, d) \leq \underset{\max}{M}(b-a) < +\infty$$

D ⤷ Partizione

- Se $P \leq Q$

$$S_s(f, P) \geq S_s(f, Q)$$

Possiamo definire

$$0 \leq S_s(f, [a,b]) := \inf_{D} S(f, D) \leq M(b-a)$$

L' idea è che

$$s_i(f,[a,b]) := \sup_{D \text{ partiz. di } [a,b]} s(f,D)$$

è la migliore sotto-approssimazione dell'area di f

- La $S_s(f,[a,b])$

è la Migliore sopra-approssimazione dell'area di f

abbiamo dim che

$$0 \leq s_i(f,D) \leq s_i(f,[a,b]) \leq S_s(f,[a,b]) \leq S_s(f,D) \leq M_{max}(b-a)$$

Riemann integrabilità

Definizione Una $f:[a,b] \to [0,+\infty)$ è (Riemann) integrabile se

$$\Rightarrow \quad s_i(f) = S_s(f)$$

In questo caso, scriviamo

$$\int_a^b f = \int_a^b \underbrace{f(x)}_{\substack{\sim m_i \\ \sim M_i}} \boxed{dx}$$

dx: Δx_i quando $(x_i - x_{i-1}) \to 0$

$\sim m_i$, $\sim M_i$ in un sottintervallo piccolo di x

Notazione

$$\int_a^b f = \int_a^b f(x)\, dx$$

x → Variabile di integrazione

Es $f:[0,1] \to [0,1]$ data per

f non integrabile

$$f(x) = \begin{cases} 1 & \text{se } x \in \mathbb{Q} \\ 0 & \text{se } x \in \mathbb{R} \setminus \{\mathbb{Q}\} \end{cases}$$

$s(f)=0$

$S(f)=1$

$s_i(f) \neq S_s(f)$

Non int

Proposiz Se $f:[a,b] \to [0,+\infty)$ è continua,

$\Rightarrow f$ è integrabile

Es
f Non cont. ma int

Proposiz Se $f: [a,b] \to \mathbb{R}$ è Riemann integrabile e

$\{D_n\} \subseteq [a,b]$ $\quad (\# D_n = n)$

è una successione di partizione tale che

$\lim_{n \to +\infty} \|D_n\| \to 0$

Allora

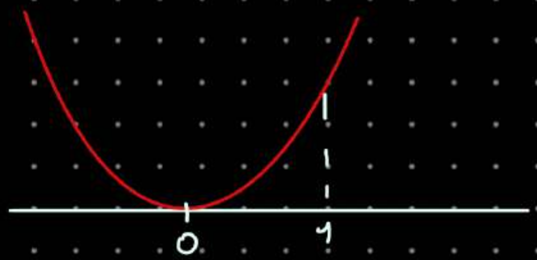
$$\int_a^b f = \lim_{n \to +\infty} \sum_{i=1}^{n} \mu_i \Delta x_i$$

con rispetto a D_n

dove $\mu_i \in \mathbb{R}$; t.c. $\mu_i \in [m_i, M_i]$

<u>Es</u> La funz $f(x) = x^2$ è integrabile su $[a,b]$

per qualsiasi $-\infty < a < b < +\infty$



Useremo una **PARTIZIONE UNIFORME** con n sottointervalli

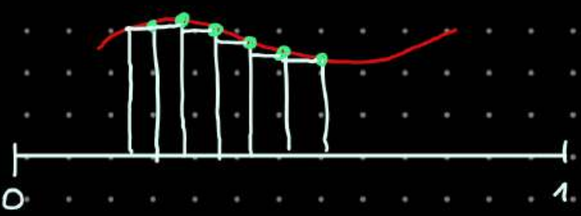
- Su $[0,1]$

$x_0 = 0$ $\qquad x_n = 1$

$x_1 = \frac{1}{n}$

$x_2 = \frac{2}{n} \quad \dots \quad x_i = \frac{i}{n}$

$$\Delta x_i = x_i - x_{i-1}$$
$$= \frac{i}{n} - \frac{i-1}{n} = \frac{1}{n}$$



Useremo i Punti destri $f(x_i)$

$$=\left(\frac{i}{n}\right)^2$$

$$=\frac{i^2}{n^2}$$

$$\sum_{i=1}^{n} \mu_i \Delta x_i = \sum_{i=1}^{n} \frac{i^2}{n^2}\cdot\frac{1}{n} = \underbrace{\frac{1}{n^3}}_{\text{e' costante}} \sum_{i=1}^{n} (i)^2 \rightarrow \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{1}{n^3} \frac{n(n+1)(2n+1)}{6}$$

Quindi (Proposiz)

$$\int_0^1 x^2\,dx = \lim_{n\to+\infty} \frac{(n+1)(2n+1)}{6n^2} =$$

$$= \lim_{n\to+\infty} \frac{2n^2+o(n^2)}{6n^2} =$$

$$= \lim_{n\to+\infty} \frac{2}{6} = \frac{1}{3}$$

se voglio calcolare fino r

$$\int_0^r x^2\,dx \overset{(?)}{=}$$

- Partiz unif

$$x_0=0, \quad x_i = i\left(\frac{r}{n}\right) \qquad x_n = r$$

- Usiamo i p.ti destri $\rightarrow \mu_i = f(x_i) = \left[i\left(\frac{r}{n}\right)\right]^2$

$$\int_0^r x^2\,dx = \lim_{n\to+\infty} \sum_{i=1}^{n} \mu_i \Delta x_i = \lim_{n\to+\infty} \sum_{i=1}^{n} i^2 \frac{r^2}{n^2}\cdot\frac{r}{n} =$$

$$= \lim_{n\to+\infty} \frac{r^3}{n^3} \sum_{i=1}^{n} i^2 = r^3 \lim_{n\to+\infty} \underbrace{\frac{n(n+1)(2n+1)}{n^3}}_{\frac{1}{3}} = \boxed{\frac{r^3}{3}}$$

OSS Definiamo

F grande $\swarrow$ $F(x) := \int_0^x y^2 \, dy = \frac{x^3}{3}$

$\frac{dF}{dx} = x^2$

In Generale, se $f : [a,b] \to [0, \infty)$ è deriv. allora

$$\int_a^b f' = f(b) - f(a)$$

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$$\int_a^b f(x)\, dx$$

$$D = \{ x_0, \dots, x_n : x_0 = a < x_1 < x_2 < \dots < x_n = b \}$$

$$s_i(f, D) = \sum_{i=0}^{n-1} \inf f \, (x_{i+1} - x_i)$$ Somma inf

S_s — sup — Somma sup

$f : [a,b] \to \mathbb{R}$ deve essere limitata

TEO $f \in C([a,b]) \Rightarrow \int_a^b f(x)\, dx$ esiste (f è integrabile su [a,b])

Teo f, g integrabile (riemann) $\Rightarrow f+g,\ f \cdot g$ sono integrabili

① $$\int_a^b (f+g)(x)\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx$$ separo la somma

② $$\int_a^b \alpha f(x)\, dx = \alpha \int_a^b f(x)\, dx$$ le costanti moltiplic. vanno fuori

$$F : f \to \int_a^b f(x)\, dx \qquad F : \mathcal{R}(a,b) \to \mathbb{R}$$

$$\mathcal{R}(a,b) = \{ f : [a,b] \to \mathbb{R} \text{ che sono integrabili} \}$$

③ $$\int_a^b f \cdot g \, dx = \int f(x) \cdot g(x) \neq \left(\int_a^b f(x)\, dx \cdot \int_a^b g(x)\, dx \right)$$

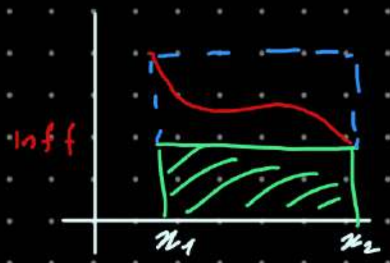
1° TEO MEDIA INTEGRALE

$f \in \mathcal{R}(a,b)$ la media int è $\frac{1}{b-a} \int_a^b f(x)\, dx = M(f)$

$-\infty < a < b < +\infty$

$$\inf_{[a,b]} f \le \frac{1}{b-a} \int_a^b f(x)\, dx \le \sup_{[a,b]} f$$

DIM



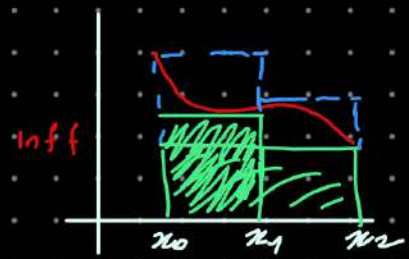
Divisione più semplice $x_0=a$ $x_1=b$

$D_0=\{x_0=a<x_1=b\}$

$s_i(f,D_0)=\inf\limits_{[a,b]} f\,(b-a)$

S_s — $\sup\limits_{[a,b]}$

se aggiungo un punto



$D_0=\{x_0=a<x_1<x_2=b\}$

$s_i(f,D_0)=\inf\limits_{[x_0,x_1]} f\,(x_1-x_0)+\inf\limits_{[x_1,x_2]} f\,(x_2-x_1)$

S_s — $\sup\limits_{[x_0,x_1]}$, $\sup\limits_{[x_1,x_2]}$

dim che usa:

(1,2) monotonia di S_s e S_i

(3) $\int_a^b f=\sup\limits_D s_i=\inf\limits_D S_s$

Corollario:

(1) $f\geq 0$ ($\leq$) integrabile su $[a,b]\Rightarrow\int_a^b f(x)\,dx\geq 0$ ($\leq$)

(2) $f\leq g$ $\quad f,g$ integ su $[a,b]\Rightarrow\int_a^b f(x)\,dx\leq\int_a^b g(x)\,dx$

$f\leq g\Rightarrow 0\leq g-f\Rightarrow 0=\int_a^b 0\,dx\leq\int_a^b(g-f)\,dx=\int_a^b g-\int_a^b f$

$\Rightarrow\int_a^b f(x)\leq\int_a^b g(x)\,dx$

(3) $\left|\int_a^b f(x)\,dx\right|\leq\int_a^b|f(x)|\,dx$ $\cancel{-|f(x)|\leq}$

Es $f\geq 0$ $\quad f\not\equiv 0$ $\quad(\exists\, x: f(x)\neq 0)$ $\quad f:[a,b]\to\mathbb{R}$ $\quad f\in\mathcal{R}[a,b]$

$\int_a^b f(x)\,dx=0$

$f(x)\begin{cases}0 & x\in[a,b]\setminus\{\alpha\}\\ 1 & x=\alpha\end{cases}$

f è integr

$s_i(f,D)=\inf\limits_{[x_0,x_1]} f+\dots+\inf\limits_{[x_{n-1},x_n]} f=0$

$S_s = 0 + 0 + 1\,(x_{k+1} - x_k) + 0 \dots$

$\forall \varepsilon > 0 \quad \exists D : |S_s(f,D) - s_i(f,D)| < \varepsilon$

$x_{k+1} - x_k < \varepsilon$

$\Rightarrow$ Scelgo D in modo che $x_{k+1} - x_k < \varepsilon$

$|S_s(f,D) - s_i(f,D)| < \varepsilon$

$\Rightarrow$ f è integrabile

Proposizione

$$\Rightarrow \int_a^b g(x) = \int_a^b f(x)$$

Es $\int_a^b f(x)\,dx$ $\quad f(x) = \begin{cases} x & [0,1) \\ e^{2x}+4 & x \in [1,2] \end{cases}$ $\quad f \in \cancel{C}[0,2]$

$$\int_0^2 f(x)\,dx = \int_0^1 f(x)\,dx + \int_1^2 f(x)\,dx = \int_0^1 x\,dx + \int_1^2 (e^{2x}+4)\,dx$$

Metodo generale per integrare funz cont. a tratti:

$f_{|[0,1]} = \begin{cases} x & x \in [0,1) \\ e^2+4 & x = 1 \end{cases}$

$\tilde{f} = x \quad \forall x \in [0,1] \qquad \int_0^1 f = \int_0^1 \tilde{f}$

Def f è continua a tratti su $[a,b]$

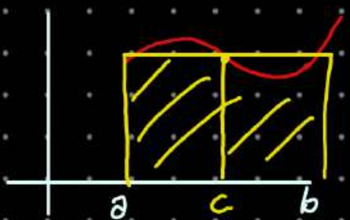
$$f \in C\left([a,b] \setminus \bigcup_{i=1}^{n} \{\alpha_i\}\right)$$

2° TEO MEDIA

Se $f \in C([a,b])$ $\quad \frac{1}{b-a}\int_a^b f(x)\,dx = f(c)$

$\exists\, c \in [a,b]$

$f(x) \geq 0$



$\int_a^b f(x)\,dx = f(c)\,(b-a)$

Area sotto al grafico — area rettangolo

$$\min_{ab} f = \mathrm{Max}_{ab} f \leq \underbrace{\frac{1}{b-a}\int_a^b f(x)\,dx}_{1° \text{ Teo Media}} \leq \sup_{ab} f = \mathrm{Max}_{ab} f$$

$$m = \min_{[a,b]} f \leq \frac{1}{b-a}\int_a^b f(x)\,dx \leq \mathrm{Max}_{[a,b]} f = M \Rightarrow$$

Teo val int mi dice che deve assum. tutti i valori $\lambda \in [m, M]$

$f \geq 0$ $\quad \int_a^b f(x)\,dx$ RAPPRESENTA L'AREA COMPRESA TRA il grafico E ASSE x

$R = \{(x,y) \in \mathbb{R}^2 \quad x \in [a,b] \quad 0 \leq y \leq f(x)\}$

il sotto grafico

$\int_a^b f(x)\,dx = \text{Area di } R$ SE $f \geq 0$ E' INTEGRABILE

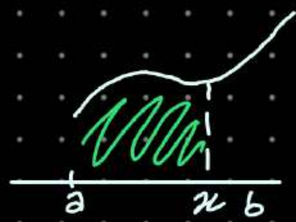
TEO FONDAMENTALE DEL CALCOLO

ipotesi $\quad f \in C([a,b])$

$\cancel{\int_a^x f(x)\,dx}$

$F(x) \overset{def}{=} \int_a^x f(t)\,dx \qquad x \in [a,b]$

Funz integrale



① F e' derivabile

$F'(x) = f(x_0) \qquad \forall x_0 \in [a,b)$

② SE $G(x)$ T.C. $G'(x) = f(x)$

$\int_a^b f(x)\,dx = G(b) - G(a) = G\Big|_a^b$

SE $G'(x) = f(x)$ ALLORA

G si dice che e' una

PRIMITIVA di f

Es $\int_0^1 x^2\,dx$

$x^2 \in C([a,b])$

$G: G'(x) = x^2 \qquad \int_0^1 x^2\,dx = G(1) - G(0)$

$G(x) = \frac{x^3}{3} + \underbrace{\pi}$ $\qquad G'(x) = x^2$ E' una PRIMIT

↳ posso mettere qualsiasi numero voglia

ES $\int_1^2 e^x = e^x\Big|_1^2 = e^2 - e = e(e-1)$

ES $\int_1^2 e^{2x} = \frac{e^{2x}}{2}\Big|_1^2 = \frac{e^4-e^2}{2}$

ES $\int_0^{\pi} \sin x \, dx = -\cos x\Big|_0^{\pi} = -\cos(\pi) - (-\cos(0))$
$= 2$

$G : G'(x) = \sin x$

$\frac{d}{dx}\cos(x) = -\sin x \qquad \frac{d}{dx}(\underbrace{-\cos x}_{\text{primit}}) = \sin x$

TABELLA INTEGRALI

Integrali indefiniti

immediati		immediati generalizzati	
$\int k\,dx = k\,x + c$	dove k è una costante	un integrale generalizzato si ottiene da un integrale immediato sostituendo $\boldsymbol{x}$ con $\boldsymbol{f(x)}$ e $\boldsymbol{dx}$ con $\boldsymbol{f'(x)\,dx}$	
$\int x^n dx = \frac{x^{n+1}}{n+1} + c$	$n \neq -1$	$\int [f(x)]^n \cdot f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1} + c$	$n \neq -1$
$\int \frac{1}{x} dx = ln\|x\| + c$		$\int \frac{f'(x)}{f(x)}\,dx = ln\|f(x)\| + c$	
$\int a^x dx = a^x\, lg_a e + c$		$\int a^{f(x)} \cdot f'(x)\,dx = a^{f(x)}\, lg_a e + c$	
$\int e^x dx = e^x + c$		$\int e^{f(x)} \cdot f'(x)\,dx = e^{f(x)} + c$	
$\int sen\,x\,dx = -\cos x + c$		$\int sen\,[f(x)] \cdot f'(x)\,dx = -\cos f(x) + c$	
$\int cos\,x\,dx = sen\,x + c$		$\int cos\,[f(x)] \cdot f'(x)\,dx = sen\,f(x) + c$	
$\int \frac{1}{cos^2 x} dx = tg\,x + c$		$\int \frac{f'(x)}{cos^2\,[f(x)]} dx = tg\,f(x) + c$	
$\int \frac{1}{sen^2 x} dx = -\,cotg\,x + c$		$\int \frac{f'(x)}{sen^2\,[f(x)]}\,dx = -\,cotg\,f(x) + c$	
$\int \frac{1}{\sqrt{1-x^2}} dx = arcsen\,x + c$		$\int \frac{f'(x)}{\sqrt{1-[f(x)]^2}}\,dx = arcsen\,f(x) + c$	
$\int \frac{1}{1+x^2} dx = arctg\,x + c$		$\int \frac{f'(x)}{1+[f(x)]^2}\,dx = arctg\,f(x) + c$	
$\int \frac{1}{\sqrt{a^2-x^2}} dx = arcsen\,\frac{x}{\|a\|} + c$		$\int \frac{f'(x)}{\sqrt{a^2-[f(x)]^2}}\,dx = arcsen\,\frac{f(x)}{\|a\|} + c$	
$\int \frac{1}{a^2+x^2} dx = \frac{1}{a} arctg\,\frac{x}{a} + c$		$\int \frac{f'(x)}{a^2+[f(x)]^2}\,dx = \frac{1}{a} arctg\,\frac{f(x)}{a} + c$	

in generale	
$\int f[g(x)] \cdot g'(x)\,dx = F\,[\,g(x)\,] + c$	l'integrale di una funzione composta $\boldsymbol{f[g(x)]}$ moltiplicata per la derivata della funzione interna $\boldsymbol{g'(x)}$ è uguale alla primitiva della funzione esterna $\boldsymbol{F[g(x)]}$

alcuni metodi di integrazione	
$\int k \cdot f(x)\,dx = k \cdot \int f(x)\,dx$	prodotto di una **costante k** per una funzione
$\int f(x) \pm g(x) \pm h(x)\,dx = \int f(x)\,dx \pm \int g(x)\,dx \pm \int h(x)\,dx$	metodo di decomposizione in **somma**

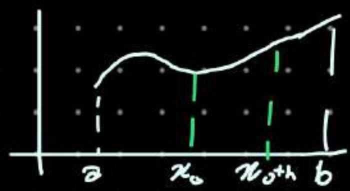
Dim TEO Fondamentale

$F(x) = \int_a^x f(t)\,dt$ è ben definita perché f e' continua

$$\lim_{x \to x_0} \frac{F(x) - F(x_0)}{x - x_0} = F'(x_0) \qquad \lim_{h \to 0} \frac{F(x_0+h) - F(x_0)}{h}$$

Per semplicita' $h > 0$
(senza perdere generalità)

$$\frac{F(x_0+h) + F(x_0)}{h} =$$



- $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx \quad \forall c \in [a,b]$
- $\int_b^a f(x)\,dx := -\int_a^b f(x)\,dx$

Proprieta'

$$\frac{F(x_0+h) + F(x_0)}{h} = \frac{\int_a^{x_0+h} f(t)\,dt - \int_a^{x_0} f(t)\,dt}{h} = \frac{\int_a^{x_0} f(t)\,dt + \int_a^{x_0+h} f(t)\,dt - \int_a^{x_0} f(t)\,dt}{h}$$

MEDIA INTEGRALE SU $[x, x_0+h]$

$$= \frac{1}{h} \int_{x_0}^{x_0+h} f(t)\,dt \qquad \left(\frac{1}{b-a} \int_a^b f(x)\,dx = f(c)\right)$$

f è cont. $\Rightarrow$ 2° MEDIA int

$$= f(c(h)) \qquad x_0 < c < x_0+h$$

$$\lim_{h \to 0} \frac{F(x_0+h) - F(x_0)}{h} = \lim_{h \to 0} f(c(h))$$

$$0 < c(h) < x_0 + h$$

TEO CARAB

$$c(h) \xrightarrow[h \to 0]{} x_0$$

$$\lim_{h \to 0} f(c(h)) = f(x_0) = \lim_{h \to 0} \frac{F(x_0+h) + F(x_0)}{h} = F'(x_0)$$

CONTINUA

② $F(x) = \int_a^x f(t)\,dt \qquad F(0) = 0 \qquad F(b) -$

G Primitiva di f $\boxed{\text{Def}}$ $G'(x) = f(x) \quad \forall x \in [a,b]$

G_1 e G_2 Primitive di f $\quad G_1'(x) = f(x)$

$$G_2'(x) = f(x)$$

$$\frac{d}{dx}\left(G_1(x)-G_2(x)\right) = G_1'(x) - G_2'(x) = f(x) - f(x) = 0 \implies G_1(x) - G_2(x) = \text{COSTANTE}$$

$$G_1(x) = G_2(x) + \text{cost}$$

PROP Se G_1 e G_2 PRIMITIVE di f su $[a,b]$

$$\implies G_1(x) = G_2(x) + \text{cost} \quad \forall x \in [a,b]$$

$f(x) = \frac{1}{x}$ $\qquad G_1(x) = \ln|x| \quad x \neq 0$

$G_1'(x) = \frac{1}{x} \quad x \neq 0$

$$G_2(x) \begin{cases} \ln x & x > 0 \\ \ln(-x) + 1 & x < 0 \end{cases}$$

$$G_2'(x) \begin{cases} \frac{1}{x} \\ \frac{1(-1)}{-x} = \frac{1}{x} & x < 0 \end{cases}$$

$$G_1'(x) = G_2'(x) = \frac{1}{x} = f(x) \quad \forall x \neq 0$$

$$G_1(x) - G_2(x) = \begin{cases} 0 & x > 0 \\ -1 & x < 0 \end{cases}$$

NON SEMPRE 2 PRIMITIVE DIFFERISCONO PER UNA COSTANTE SE IL DOMINIO NON È CONNESSO

Se $G(x)$: $G'(x) = f(x)$

$$G'(x) = F'(x) = f(x) \quad \forall x \in [a,b]$$

$$G(x) = F(x) + c \quad \forall x \in [a,b]$$

$$G(x)\Big|_a^b = G(b) - G(a) = F(x) + c\Big|_a^b = F(b) + c - F(a) + c = \int_a^b f(t)\,dt + c - c$$

$$\overset{!}{=} \int_a^b f(t)\,dt$$

Spiega perché $\int_a^b f(x)\,dx = G\Big|_a^b$ se $f \in \mathcal{R}[a,b]$

Formula di Torr - Barrow

ES $\displaystyle\int_{-\pi}^{\frac{\pi}{2}} (x + x^4 - 10x^{21} + \sin x)\,dx = \int_{-\pi}^{\frac{\pi}{2}} \frac{x^2}{2}\,dx + \ldots$

(Formulario) $\displaystyle\int_a^b \sum_{i=1}^{n} \alpha_i f_i = \sum_{i=1}^{n} \alpha_i \int_a^b f_i(x)\,dx$

! $\displaystyle\int_a^b (f \cdot g)(x)\,dx \neq \int_a^b f(x)\,dx \cdot \int_a^b g(x)\,dx$

ES $\displaystyle\int_2^5 e^{x^2} \cdot x\,dx \qquad f(x) = e^{x^2} \qquad g(x) = x$

$$\int_2^5 e^{x^2} \cdot x\,dx = \int_2^5 \boxed{\frac{d}{dx}\left(\frac{e^{x^2}}{2}\right)}\,dx = \frac{e^{x^2}}{2}\Big|_2^5 = \frac{e^{25} - e^4}{2}$$

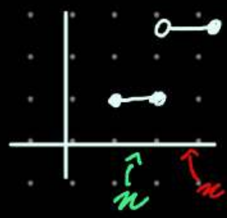
$$f(x) = \frac{d}{dx}\left(\frac{e^{x^2}}{2}\right)$$

$$F' = f \qquad F(x) = \frac{e^{x^2}}{2}$$

OSS $\int_a^b f'(x)\,dx = f\Big|_a^b = f(b)-f(a)$

↳ $\int_a^b \frac{df}{\cancel{dx}}\,\cancel{dx} = f(b)-f(a)$

$$f(x)=\begin{cases}1 & x\in[2,3]\\ 5 & x\in(3,4]\end{cases}$$



$f:[3,4]\to\mathbb{R}$

Calcolare $F(x)=\int_2^x f(t)\,dt \qquad x\in[2,4]$

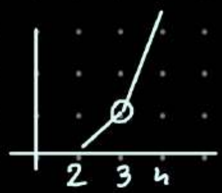
$$F(x)=\int_2^x f(t)\,dt = \int_2^x 1\,dt \qquad 2\le x\le 3$$

$$= t\Big|_2^x = x-2$$

$$F(x)=\int_2^x f(t)\,dt = \int_2^3 f(t)\,dt + \int_3^x f(t)\,dt = \int_2^3 1\,dt + \int_3^x 5\,dt =$$

$$= t\Big|_2^3 + 5\,t\Big|_3^x \qquad = 1+5(x-3)$$

$F(x)$



$F'(3) = \text{NE}$

$F'_-(3)=1 \neq F'_+(3)=5$

25/11/25

$$\int_a^b f(x)\,dx = G\Big|_a^b = G(a)-G(b)$$

$G'(x) = f(x)$

G è una PRIMITIVA di f

f è continua a tratti, $f \subset ([a,b] \setminus \bigcup_i^n \{\alpha_i\})$, $\lim_{x\to\alpha_i^\pm} f(x)$ esiste finito

Integrazione Per Parti

$$\int_a^b \frac{df}{dx}\,dx = f(b)-f(a) = \int_a^b f'(x)\,dx \qquad f' \subset [a,b]$$

$$\int_a^b \frac{df}{dx}(f\cdot g)\,dx = fg\Big|_a^b = f(b)g(b) - f(a)g(a)$$

$$\int_a^b f'(x)\cdot g(x) + f(x)\,g'(x)\,dx$$

FORMULA INTEG PARTI

- $$\int_a^b f'(x)\,g(x)\,dx = -\int_a^b f(x)\,g'(x)\,dx + f\cdot g\Big|_a^b$$

ES
$$\int_0^\pi \underset{f'}{x}\,\underset{g}{\sin(x)}\,dx = \frac{x^2}{2}\sin(x)\Big|_0^\pi - \int_0^\pi \frac{x^2}{2}\cos x\,dx$$ strada sbagliata

$$\int_0^\pi \underset{g}{x}\,\underset{f'}{\sin x} = -x\cos x\Big|_0^\pi - \int_0^\pi -\cos(x)\cdot 1\,dx$$
$$= -x\cos x\Big|_0^\pi + \sin x\Big|_0^\pi = \sin x - x\cos x\Big|_0^\pi$$

ES
$$\int_0^1 \underset{g}{(x^2+3x)}\,\underset{f'}{\cos x}\,dx = (x^2+3x)\sin x\Big|_0^1 - \int_0^1 (2x+3)\sin x\,dx$$
$$= (x^2+3x)\sin(x)\Big|_0^1 - 3\int_0^1 \sin x\,dx - 2\int_0^1 x\sin x\,dx$$
$$= (x^2+3x)\sin(x) + 3\cos x\Big|_0^1 - 2\left[\sin x - x\cos x\Big|_0^1\right]$$

grado $P_n = n$

- $$\int_a^b \underset{g}{P_n(x)} \, \underset{f'}{\sin(x)} \, dx = \qquad P_n(x) = \sum_{k=0}^{n} a_k x^k \quad a_k \in \mathbb{R} \neq 0$$

$$= -P_n(x) \cos(x) \Big|_a^b + \int_a^b (P_n'(x) \cos(x) \, dx$$

grado $P_n' = n-1$

- $$\int_1^{e^2} \underset{g}{x} \, \underset{f'}{e^x} \, dx = x e^x \Big|_1^{e^2} - \int_1^{e^2} 1 e^x = x e^x - e^x \Big|_1^{e^2} = e^2 e^{e^2} - e^{e^2} - (1 \cdot e - e)$$

$$\int_a^b \underset{g}{P_n(x)} \, \underset{f'}{e^x} \, dx = P_n(x) e^x \Big|_a^b - \int_a^b P_n'(x) e^x \, dx$$

-

$$\int_a^b \underset{g}{P_n(x)} \, \underset{f'}{e^{\alpha x}} \, dx = P_n(x) \frac{e^{\alpha x}}{\alpha} \Big|_a^b - \int_a^b \frac{P_n'(x)}{\alpha} e^{\alpha x} \, dx$$

$0 < a < b$

$$\int_a^b \ln x \, dx = \int_a^b \ln(x) \cdot 1 \, dx = x \ln x \Big|_a^b - \int_a^b x \frac{1}{x} \, dx$$

$$= x \ln x \Big|_a^b - x \Big|_a^b =$$

$$= x \ln x - x \Big|_a^b$$

$$\int_a^b \underset{f'}{x} \, \underset{g}{\ln(x)} \, dx = \frac{x^2}{2} \ln x \Big|_a^b - \int_a^b \frac{x^2}{2} \frac{1}{x} \, dx =$$

$$= \frac{x^2}{2} \ln x \Big|_a^b - \frac{x^2}{4} \Big|_a^b = \frac{x^2}{2} (\ln x - \frac{1}{2} \Big|_a^b$$

- $$\int_a^b P_n(x) \ln x \, dx = \widetilde{P_n} \ln x \Big|_a^b - \int_a^b \frac{\widetilde{P_n}(x)}{x} \, dx$$

$$(\widetilde{P_n}(x))' = P_n(x)$$

$$\int_0^1 \arctan x \, dx = \int_0^1 \underset{f'}{1} \cdot \underset{g}{\arctan x} \, dx = x \arctan x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} dx$$

$$= x \arctan x \Big|_0^1 - \int_0^1 \frac{1}{2} \frac{2x}{(1+x^2)} dx =$$

$$= \text{arc}$$

2° ora

$$\int_a^b \sin^2 x \, dx$$

$$\int_a^b \sin^2 x \cos x \, dx = \frac{\sin^3 x}{x} \Big|_a^b$$

$$\int_a^b f^2_{(x)} \, f'(x) \, dx = \int_a^b \frac{d}{dx} \frac{f^3(x)}{3} dx = \frac{f^3(x)}{3} \Big|_a^b$$

$$\int_a^b f^n_{(x)} \, f'(x) \, dx = \frac{f^{n+1}(x)}{n+1} \Big|_a^b$$

$$\int_a^b \underset{f'}{\sin(x)} \cdot \underset{g}{\sin(x)} \, dx = \sin(x) (-\cos(x)) \Big|_a^b - \int_a^b \cos x (-\cos x) \, dx$$

$$= -\sin x \cos x \Big|_a^b + \int_a^b \cos^2(x) \, dx$$

$\sin^2 + \cos^2 = 1 \quad \Rightarrow \cos^2 = 1 - \sin^2 x$

$$\boxed{\int_a^b \sin^2 x \, dx} = -\sin(x)\cos(x) \Big|_a^b + \int_a^b (1 - \sin^2 x) \, dx$$

$$= \boxed{-\sin x \cos x \Big|_a^b + x \Big|_a^b} \boxed{-\int_a^b \sin^2 x \, dx}$$

$$\frac{2 \int_a^b \sin^2 x}{2} = \frac{x - \sin x \cos x \Big|_a^b}{2}$$

$$\int_a^b \sin^2 x \, dx = \frac{x - \sin x \cos x}{2}$$

- $\int_0^{\pi} \underset{f'}{\sin x} \underset{g}{\cos x}\, dx = \frac{1}{2}\int_0^{\pi} 2\sin x \cos x\, dx = \frac{1}{2}\int_0^{\pi} \sin(2x)\, dx$

$= \frac{1}{2}\left(-\frac{\cos 2x}{2}\right)\Big|_0^{\pi} = -\frac{1}{4}\cos 2x\Big|_0^{\pi}$

- $\int_0^{\frac{\pi}{2}} \sin x \sin(3x)\, dx$

$e^{inx} = \cos(nx) + i\sin(nx)$

$(e^{ix})^n = (\cos(x) + i\sin x)^n$

$e^{i3x} = \cos(3x) + i\,\mathrm{sen}(3x) =$

$= (\cos x + i\,\mathrm{sen}\,x) =$

$= \cos^3 x + 3\cos^2(x)\cdot i\sin(x) + 3\cos(x)(i\sin(x))^2 + (i\sin(x))^3$

$= \cos^3(x) - 3\cos(x)\sin^2(x) + i\left[3\cos^2(x)\sin(x) - \sin^3(x)\right]$

- $\int_0^{\frac{\pi}{2}} \underset{f'}{\sin(x)} \cdot \underset{g}{\sin(3x)}\, dx = \underset{f}{-\cos(x)}\, \underset{g}{\sin(3x)}\Big|_0^{\pi/2} - 3\int_0^{\frac{\pi}{2}} \underset{f}{-\cos(x)} \cos(3x)\, dx$

$= -\cos(x)\sin(3x)\Big|_0^{\pi/2} + 3\int_0^{\pi/2} \underset{f'}{\cos x}\, \underset{g}{\cos(3x)}\, dx$

$= -\cos(x)\cdot\sin(3x)\Big|_0^{\pi/2} + 3\left[\sin x\cos(3x)\Big|_0^{\pi/2} + 3\int_0^{\pi/2} \sin x \sin(3x)\right]$

$\int_0^{\frac{\pi}{2}} \sin x = -\cos(x)\sin 3x + 3\sin x\cos 3x\Big|_0^{\pi/2} + 9\int_0^{\pi/2} \sin x \sin 3x\, dx$

Trovare una primitiva di $e^x \sin x$

$\int_a^b \underset{f'}{e^x}\, \underset{g}{\sin x}\, dx = \underset{f}{e^x}\, \underset{g}{\sin x}\Big|_a^b - \int_a^b \underset{f}{e^x}\, \underset{g'}{\cos x}\, dx$

$= e^x \sin x\Big|_a^b - \left[e^x\cos(x)\Big|_a^b + \int_a^b e^x(+\sin x)\, dx\right]$

$= e^x\sin x\Big|_a^b - e^x\cos x\Big|_a^b - \underline{\int_a^b e^x \sin x\, dx}$

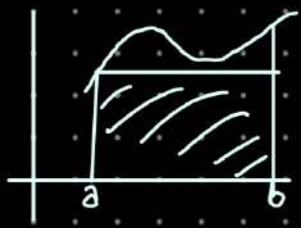
$\frac{2}{2}\int_a^b e^x \sin x\, dx = \frac{e^x(\sin x - \cos x\Big|_a^b}{2}$

$''G''$

Integrale indefinito

$\int e^x \sin x\, dx$

Tutte le primitive di $e^x \sin x$



$f \in C^1$

$\int_0^b f(x)dx$

$f(a)(b-a)$

$f(x) \equiv f(a)$

$\cancel{f(x)dx = f(a)(b-}$

$f(x) = f(a) + f'(c)(x-a)$

F Lagrange ordine 0

$f, f' \in C([a b])$

$|f'| \leq C = \|f'\|_\infty$ (?)

$$\int_a^b f(x)\,dx = \int_a^b f(a)\,dx + \int_a^b f'(c)(x-a)\,dx$$

$$\int_a^b f(x)\,dx - \int_a^b f(a)\,dx = \int_a^b f'(c)(x-a)\,dx$$

$$\left|\int_a^b f(x)\,dx - f(a)(b-a)\right| = \left|\int_a^b f'(c)(x-a)\,dx\right| \leq \int_a^b |f'(c)|$$

$f \leq g \Rightarrow \int f \leq \int g$

$$= \|f'\|_\infty \int_a^b (x-a)\,dx$$

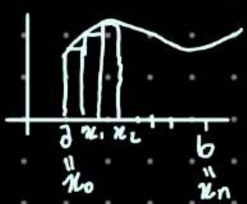
$$= \|f'\|_\infty \left.\frac{(x-a)^2}{2}\right|_a^b = \|f'\|_\infty \left[\frac{(b-a)^2-(a-a)^2}{2}\right]$$

$$= \|f'\|_\infty \frac{(b-a)^2}{2}$$

$\int x\,dx = \frac{x^2}{2}$

$\int$

$|f'| << 1$



(Metodo dei rettangoli)

$$\int_a^b f(x)\,dx \qquad \sum_{i=0}^{n-1} f(x_i)(x_{i+1}-x_i)$$

Cambio di variabile

idea: invece di integrare in "dx" (variabile x), possiamo riscrivere l'integrale in una variabile "dt" più semplice

Requisiti:

1) Come si trasforma la DIFFERENZIALE
$dx \rightsquigarrow dt$

2) Come si trasformano gli estremi
$\int$

ES Per calcolare
$$\int_a^b f(x)\, dx$$
facciamo il cambio di variabile
$$x = g(t)$$

dove $g: \mathbb{R} \to \mathbb{R}$ invertibile e Biettiva
$([a,b] \to [c,d])$

ALLORA:

- DIFFERENZIALE
$$dx = d(g(t)) = g'(t)\, dt$$

- ESTREMI $\int_a^b f$

se $x = a \rightsquigarrow t = g(a)$

se $x = b \rightsquigarrow t = g(b)$!

REGOLA:

$x = g(t)$
$dx =$

$$\int_a^b f(x)\, dx = \int_{g(a)}^{g(b)} f(g(t))\, g'(t)\, dt$$

- Necessario scegliere g che semplifichi la f integrale

NOTA BENE

$f^{-1}(x) \neq f(x)^{-1}$

$f^{-1}(x)$: Simbolo che indica la funz. inversa

$f(x)^{-1}$: NUMERO elevato alla -1 $(f(x))^{-1}$

ES SOSTITUZIONE LINEARE

Calcola $\int_2^5 (3x-1)\, dx$

$t = 3x - 1$

$x = \frac{t+1}{3}$

$= g(t) \rightarrow g'(t) = \frac{1}{3}$

$\int_5^{14} t \, \underbrace{\tfrac{1}{3}\, dt}_{dx} =$

$= \frac{1}{3} \frac{t^2}{2} \Big|_5^{14} = \frac{1}{6}(14^2 - 5^2)$

$2 \rightarrow g^{-1}(2)$
$= 3(2) - 1$
$= 5$

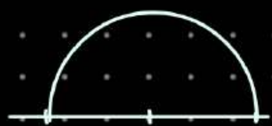
$5 \rightarrow g^{-1}(5)$
$= 3(5) - 1$
$= 14$

ES

$\int_2^5 (3x-1)^n \, dx = \int_5^{14} t^n \frac{1}{3}\, dt$ $\quad t = 3x - 1$

ES NOTEVOLE

$\int_{-1}^{1} \sqrt{1 - x^2}\, dx =$



1) • Sostituzione

$x = \sin(t)$

• differenziare

$dx = \sin(t)'\, dt$

$\cos(t)\, dt$

• Estremi

$x = \pm 1 \rightarrow t = \sin^{-1}(\pm 1)$

$= \arcsin(\pm 1)$

$= \pm \frac{1}{2}\pi$

$$\int_{-1}^{1} \sqrt{1-x^2}\, dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{1-\sin^2(t)} \cdot \cos t \, dt$$

$x = \sin t$
$dx = \cos t \, dt$

$\cos t \geq 0 \quad \forall\, t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos t \cdot \cos t \, dt$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 t$$

$\cos^2 t = \frac{1+\cos(2t)}{2}$

$$= \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1+\cos 2t)\, dt =$$

$$= \underbrace{\frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 \, dt}_{} + \frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(2t)\, dt$$

$$= \frac{\pi}{2} + \frac{1}{2} \left(\frac{\sin 2t}{2}\right)\Bigg|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$\left(\frac{\sin(2t)}{2}\right)' = \frac{\cos 2t \cdot \cancel{2}}{2}$
$= \cos 2t$

TFC $\int_a^b f' = f(b) - f(a) = f\Big|_a^b$

$$= \frac{\pi}{2} + \frac{1}{2}\, \cancel{\frac{\sin(\pi) - \sin(-\pi)}{2}}^{\,0}$$

$$= \frac{\pi}{2}$$

Metodo alternativo (integr per parti) $\int_{-1}^{1} \sqrt{1-x^2}\, dx$

Ricordo: $\int_a^b u \cdot v' \, dx = -\int_a^b u' v \, dx + uv\Big|_a^b$

Poniamo $u(x) = \sqrt{1-x^2}$

$v'(x)\, dx = 1 \cdot dx$

$\Rightarrow v(x) = x$

$v'(x)\, dx = 1 \cdot dx$

$$\int_{-1}^{1} \sqrt{1-x^2}\, dx = \int_{-1}^{1} \sqrt{1-x^2} \cdot 1 \, dx$$

$$= \int_{-1}^{1} u(x)\, v'(x)\, dx$$

Int p.p. $= -\int_{-1}^{1} u'(x)\, v(x)\, dx + uv\Big|_{-1}^{1} =$

$u'(x) = \frac{d}{dx}\left(\sqrt{1-x^2}\right)(x)$

$= -\frac{x}{\sqrt{1-x^2}} = -\int_{-1}^{1} \overbrace{\left(-\frac{x}{\sqrt{1-x^2}}\right)}^{u'(x)} x\, dx + \cancel{\left(x\sqrt{1-x^2}\right)}\Big|_{-1}^{1}$

$= \int_{-1}^{1} \frac{x^2}{\sqrt{1-x^2}}\, dx + 0$

$\frac{x^2}{\sqrt{1-x^2}} = \frac{1-(1-x^2)}{\sqrt{1-x^2}} = \frac{1}{\sqrt{1-x^2}} - \sqrt{1-x^2} =$

$= \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\, dx - \underline{\underline{\int_{-1}^{1} \sqrt{1-x^2}\, dx}}$

Quindi

$\int_{-1}^{1} \sqrt{1-x^2}\, dx = \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\, dx - \int_{-1}^{1} \sqrt{1-x^2}\, dx$

$2\int_{-1}^{1} \sqrt{1-x^2}\, dx = \frac{1}{2}\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\, dx$ $= \arcsin'(x)$

$= \frac{1}{2}\int_{-1}^{1} \arcsin'(x)\, dx =$

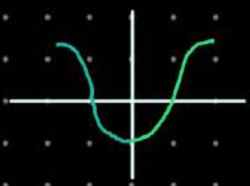
T.F.C. $= \frac{1}{2} \arcsin x\Big|_{-1}^{1} = \frac{1}{2}\left(\arcsin(1) - \arcsin(-1)\right)$

$= \frac{1}{2}\left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right) = \frac{1}{2}\pi$

Funz pari e dispari

Ricordiamo

- f e' pari se
  $f(x) = f(-x)$



- f e' dispari se
  $f(-x) = -f(x)$

- Se f e' pari

$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx = 2\int_{-a}^{0} f(x)\,dx$$

- Se f e' dispari

$$\int_{-a}^{a} f(x)\,dx = 0$$

ES

$$\int_{-3}^{3} x^3 \overset{disp \Rightarrow}{\cos(x)}\,dx = 0$$

- x^3 dispari
- $\cos(x)$ pari

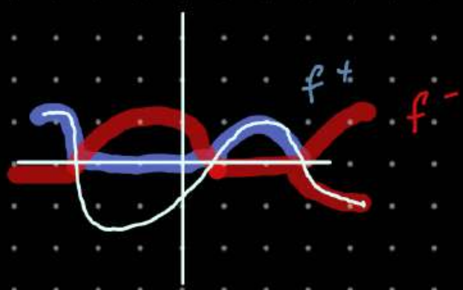
$\rightarrow$ prodotto $P \times D$ = Dispari

Integ funz Reali

Per una funz $f: [a,b] \rightarrow \mathbb{R}$ def

$f^+ := \max\{f, 0\}$ (≥ 0)

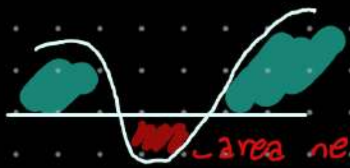
$f^- := \max\{-f, 0\}$ (≥ 0)



- $f^+ > 0 \Rightarrow f^- = 0$
  $f^- > 0 \Rightarrow f^+ = 0$
- $f^+ + f^- = |f|$
- $f = f^+ - f^- =$

DEFiniamo $\int_a^b f := \int_a^b f^+ - \int_a^b f^-$

$$\int_a^b f^+ + \int_a^b f^- = \int_a^b |f|$$

- $|f|$ é integr. $\iff$ f^-, f^+ sono integ
- def: $f:[a,b] \to \mathbb{R}$ e integ $\iff$ $|f|:[a,b] \to [0,+\infty)$ e' integ
- $a\int f = \int af \qquad a \in \mathbb{R}$
- $\int (f+g) = \int f + \int g$;
- $\left|\int_a^b f\right| \leq \int_a^b |f|$ Disuguagl. Triangolare
- 

Cambio di variabile Trigonometrici

$f(x) = \sqrt{a^2 - x^2} \qquad x \in [a,b]$

$x = \sin(t)$

- $f(x) = \dfrac{1}{x^2+a^2}$

$x = \arctan(t\, a)$

Integ fun razionali $\left(f(x) = \dfrac{P(x)}{Q(x)} \right)$ con P, Q polin.

consideriamo

Q di grado ≤ 2

coeff non 0 del monomio piu' grande

$f(x) = \dfrac{P(x)}{Q(x)}$, grado$(Q) \leq 2$

costante

- Se grado$(Q) = 0 \to Q(x) = C_0 \in \mathbb{R}$

$f(x) = C_0^{-1} P(x)$ e' un Polin.

- Se grado$(Q) = 1$

$\longrightarrow Q(x) = ax + b, \quad a, b \in \mathbb{R}$

consideriamo SOLO integr. dove f é continua:

$$\int_A^B \frac{P(x)}{ax+b} \; ; \quad \text{dove} \quad |ax+b| > 0 \quad \text{su } [A,B]$$

Algoritmo della divis (Polinomi)

Se $P(x)$, $Q(x)$ polin grado $(P) \geq$ grado (Q)

allora $P(x) = S(x)\,Q(x) + R(x)$

dove

- S, R sono polin
- grado (S) = grado (P) - grado (Q)
- grado (R) < grado (Q)

$$f(x) = \frac{P(x)}{Q(x)} = \frac{S(x)\,Q(x) + R(x)}{Q(x)}$$

$$= S(x) + \frac{R(x)}{Q(x)}$$

Quindi $\displaystyle\int_A^B f(x) = \int_A^B S(x) + \int_A^B \frac{R(x)}{Q(x)}$ — $g(R) < g(Q)$

1 h di Appunti

• Area  $=$ $2 \cdot$ Area 

• $\sum_{i=1}^{n}$ Area  $= \sum_{i=1}^{n} 1 \cdot \sin\left(\frac{\theta}{2}\right)$

$$= n \cdot \sin\left(\frac{\theta}{2}\right)$$

$\theta = \frac{\pi}{n}$

$$= n \cdot \sin\left(\frac{\pi}{2n}\right)$$

$\lim_{n \to +\infty} \sum_{i=1}^{n}$ area  $= \lim_{n \to +\infty} n \sin\left(\frac{\pi}{2n}\right)$

$$= \lim_{n \to \infty} \frac{\pi}{2} \cdot \frac{2}{\pi} \cdot n \cdot \sin\left(\frac{\pi}{2n}\right)$$

$$= \frac{\pi}{2} \lim_{n \to +\infty} \frac{\sin\cancel{\left(\frac{\pi}{2n}\right)}}{\cancel{\left(\frac{\pi}{2n}\right)}} \quad 1$$

$$= \frac{\pi}{2}$$

28/11

Metodo dei Rettangoli

$$\int_1^2 \sin(e^{x^2})\,dx$$

$x_0 = a \qquad \Delta x = \dfrac{b-a}{n}$

$x_k = x_0 + k\,\dfrac{b-a}{n} \qquad \begin{array}{ll} k=0 & x_0 \\ k=n & x_n \end{array}$

$$S = \sum_{k=0}^{n-1} f(x_k)\,(x_{k+1} - x_k)$$

$$\longrightarrow \qquad \left| \int_a^b f(x)\,dx - \sum_{k=0}^{n-1} f(x_k)\,(x_{k+1}-x_k) \right| \leq \text{piccola}$$

$$f(x) = f(x_k) + f'(y_k)(x - x_k) \qquad x_k < x < x_{k+1}$$

Uso Lagrange su $[x_k, x_{k+1}]$ $\qquad x_k < y_k < x_{k+1}$

$$\int_a^b f(x)\,dx = \sum_{k=0}^{n-1} \int_{x_k}^{x_{k+1}} f(x)\,dx = \sum_{k=0}^{n-1} \int_{x_k}^{x_{k+1}} [f(x_k) + f'(y_k)(x - x_k)]\,dx$$

$$= \sum_{k=0}^{n-1} \left[\int_{x_k}^{x_{k+1}} f(x_k)\,dx + \int_{x_k}^{x_{k+1}} f'(y_k)(x-x_k)\,dx \right]$$

$$= \sum_{k=0}^{n-1} \underbrace{f(x_k)(x_{k+1}-x_k)}_{\text{numero}} + \underbrace{\sum_{k=0}^{n-1} \int_{x_k}^{x_{k+1}} f'(y_k)(x-x_k)\,dx}_{\text{Resto}}$$

$$\left| \int_a^b f(x)\,dx - \sum_{k=0}^{n-1} (f(x_k)(x_{k+1}-x_k) \right| = \left| \sum_{k=0}^{n-1} \int_{x_k}^{x_{k+1}} f'(y_k)(x-x_k)\,dx \right|$$

Uso Disug. triangolare

$$\leq \sum_{k=0}^{n-1} \left| \int_{x_k}^{x_{k+1}} f'(y_k) \cdot |x - x_k|\,dx \right.$$

$$\left| \int_a^b f \right| \leq \int_a^b |f|$$

$$\leq \sum_{k=0}^{n-1} \int_{x_k}^{x_{k+1}} |f'(y_k)| \cdot |x - x_k|\,dx$$

Supponiamo

$\max |f'(x)| \leq M$

$x \in [a,b]$

$$\sum_{k=0}^{n-1} \int_{x_k}^{x_{k+1}} M(x-x_0)\, dx$$

$$\sum_{k=0}^{n-1} M \int_{x_k}^{x_{k+1}} (x-x_k)\, dx = M \sum_{k=0}^{n-1} \left[\frac{(x-x_k)^2}{2} \Bigg|_{x_k}^{x_{k+1}} \right]$$

$|x-x_k| = (x-x_k) \Leftarrow x_k < x$

$x_k \quad x_{k+1}$

$$M \sum_{k=0}^{n-1} \left[\frac{(x_{k+1}-x_k)^2}{2} - \frac{(x_k-x_k)^2}{2} \right]$$

$$= M \sum_{k=0}^{n-1} \frac{1}{2}\left(\frac{b-a}{n}\right)^2 = \frac{M}{2} n \left(\frac{b-a}{n}\right)^2 = \frac{M}{2}(b-a)^2 \frac{1}{n}$$

$x_{k+1} - x_k = \frac{b-a}{n}$

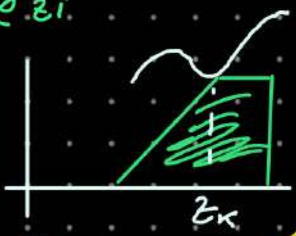
$$\int_a^b f(x)\, dx - \sum f(x_k)(x_{k+1}-x_k) \leq \frac{M}{2}(b-a)^2$$

$$|f'(x)| = |\cos e^{x^2} \cdot e^{x^2} \cdot 2x| \leq 1 e^4 \cdot 4$$

$n > 10^2 \frac{e^4 \cdot 4}{2}$

Provo il metodo dei trapezi

$$\sum_{k=0}^{n-1} f(z_k)(x_{k+1}-x_k)$$



$z_k = \frac{x_k + x_{k+1}}{2}$

$f(x) = f(z_k) + f'(z_k)(x-z_k) + f''(z_k)\frac{(x-z_k)^2}{2}$

t.y. all'ordine 2 a pt z_k

$$\int_{x_k}^{x_{k+1}} f(x)\, dx = \int_{x_k}^{x_{k+1}} \quad dx$$

$\max |f''(y)| \leq M_2$

$y \in (a,b)$

$$= \frac{M_2}{24}(b-a)^3 \frac{1}{n^2}$$

quando ho $\int_a^b f(x)\, dx$

Somma inf $\quad s = \sum \inf f(x)(x_{k+1}-x_k)$

Vediamo l'integrale Generalizzato su $[a, +\infty)$

$\int_a^{\infty}$ $\quad f:[a,+\infty) \to \mathbb{R} \quad f_{|[a,b]} \in C([a,+\infty))$

$\Rightarrow \forall\, b \geq a \quad \exists \int_a^b f(x)\,dx$

$\lim_{b\to\infty} \left(\int_a^b f(x)\,dx \right)$

$f(x) = \frac{1}{x^2} \quad [a,\infty) = [1,+\infty)$

$$\int_a^b \frac{1}{x^2}\,dx = \int_a^b x^{-2}\,dx = \frac{1}{1-2} x^{1-2} \Big|_1^b = -\frac{1}{x}\Big|_1^b = -\frac{1}{b}+1$$

$$\lim_{b\to+\infty} \left[\int_1^b \frac{1}{x^2} \right] = \lim_{b\to+\infty} \left[1 - \frac{1}{b} \right] = 1 \stackrel{def}{=} \int_1^{\infty} \frac{1}{x^2}\,dx$$

2° ora

DEF $\quad f:[a,+\infty) \to \mathbb{R} \quad f$ integr. su $[a b]$ $\forall b > a$

$$\int_a^{+\infty} f(x)\,dx = \lim_{b\to+\infty} \left[\int_a^b f(x)\,dx \right.$$ se il Lim esiste Finito

Esempio

$\bullet \int_0^{+\infty} e^{-ax} := \lim_{b\to+\infty} \int_0^b e^{-ax}\,dx$

$$\int_0^b e^{-ax}\,dx = \frac{e^{-ax}}{-a}\Big|_0^b = \frac{e^{-ab}-1}{-a} \qquad a \neq 0$$

$$\int_0^b e^{-ax}\,dx = \int_0^b 1 = b$$

Se $a=0$ $\quad \underset{N.E}{\overset{\shortparallel}{\int_0^{+\infty} e^{-ax}\,dx}} = \lim_{b\to+\infty} b = +\infty$

Se $a \neq 0$ $\quad \lim_{b\to+\infty} \left(\int_0^b e^{-ax}\,dx \right) = \lim_{b\to+\infty} \frac{e^{-ax}-1}{-a} \sim$

$\bullet$ ES IMPORTANTE

$$\int_1^{+\infty} \frac{1}{x^\beta}\,dx = \lim_{b\to+\infty} \left[\int_1^b \frac{1}{x^\beta}\,dx \right]$$

$$\int_1^b \frac{1}{x^\beta}\,dx = \begin{cases} \beta = 1 & = \ln b \\ \beta \neq 1 & \int_1^b x^{-\beta}\,dx = \frac{1}{1-\beta} x^{1-\beta}\Big|_1^b = \frac{1}{1-\beta}\left(b^{1-\beta}-1\right) \end{cases}$$

$$\lim_{b\to+\infty}\int_1^b \frac{1}{x^\beta}\,dx = \begin{cases} \beta=1 & \lim_{b\to+\infty} \ln b = +\infty \\ \beta\neq 1 & \lim_{b\to+\infty}\frac{1}{1-\beta}\left(b^{1-\beta}-1\right) = \begin{cases} +\infty & 1-b>0 \\ \frac{1}{b-1} & 1-b<0 \end{cases} \end{cases}$$

$$\boxed{\int_1^{\infty}\frac{1}{x^\beta}\,dx = \frac{1}{b-1} \quad \text{solo se } \beta>1}$$

Es

- $$\int_0^{+\infty}\frac{1}{1+x^2}\,dx = \lim_{b\to+\infty}\left[\int_0^b \frac{1}{1+x^2}\,dx\right] = \lim_{b\to+\infty}\left[\arctan\right.$$

$$= \lim_{b\to+\infty}(\arctan b - \arctan 0) = \frac{\pi}{2}$$

Es

$$\int_2^{+\infty}\frac{1}{x(x-1)}\,dx \qquad \in C([2,+\infty))$$

$$\int_2^b \frac{1}{x(x-1)}$$

$$\frac{1}{x(x-1)} = \frac{A}{x}+\frac{B}{x-1} = -\frac{1}{x}+\frac{1}{x-1}$$

$$\frac{A}{x}+\frac{B}{x+1} \qquad \frac{A(x-1)+Bx}{x(x-1)} = \frac{1}{x(x-1)}$$

$$\frac{(A+B)x-A}{x(x-1)} = \frac{1}{x(x-1)}$$

$$\begin{cases} A+B=0 \\ -A=1 \end{cases} \qquad \begin{array}{l} A=-1 \\ B=1 \end{array}$$

$$\int_2^b -\frac{1}{x}\,dx + \int_2^b \frac{1}{x-1}\,dx =$$

$$= -\ln|x| + \ln|x-1|\Big|_2^b = \ln\frac{b-1}{b} + \ln 2$$

$$\lim_{b\to+\infty}\int_2^b \frac{1}{x(x-1)}\,dx = \lim_{b\to+\infty}\left(\ln\left(\frac{b-1}{b}\right)+\ln(2)\right) = \ln 2$$

ALTRO MODO

$$\frac{1}{x(x-1)}$$

Se Denom ≥ 2

$$\int_2^b \frac{1}{x^2(x-1)}\,dx \qquad \frac{A}{x}+\frac{B}{x^2}+\frac{C}{x-1}=\frac{1}{x^2(x-1)}$$ cerco A,B,C

Se avessi avuto

$$\frac{1}{x^4(x-1)^2} \qquad \frac{A}{x}+\frac{B}{x^2}+\frac{C}{x^3}+\frac{D}{x^4}+\frac{E}{(x-1)}+\frac{F}{(x-1)^2}=\frac{1}{x^4(x-1)^2}$$

① Modo

$$\frac{A}{x}+\frac{B}{x^2}+\frac{C}{x-1}=\frac{1}{x^2(x-1)}=\frac{Ax(x-1)+B(x-1)+Cx^2}{x^2(x-1)}$$

$$=\frac{(A+C)x^2+(B-A)x-B}{x^2(x-1)}$$

$$\begin{cases} -B=1 \\ B-A=0 \\ A+C=0 \end{cases}$$

② Modo

$\boxed{1}$ $$(x-1)\left(\frac{A}{x}+\frac{B}{x^2}+\frac{C}{x-1}\right)=\frac{1}{x^2(x-1)}(x-1) \quad \forall x\neq 0,1$$

$[x-x_0]$

$$C \xleftarrow{x\to 1} \frac{A(x-1)}{x}+\frac{B(x-1)}{x^2}+C=\frac{1}{x^2}\xrightarrow{x\to 1} 1$$ Faccio lim $x\to 1$

$\hookrightarrow C=1$

Si fa il Lim con $x\to x_0$

$(x-0)$

$\boxed{2}$ $$x\left(\frac{A}{x}+\frac{B}{x^2}+\frac{C}{x-1}\right)=\frac{1}{x^2(x-1)}x$$

NON VA $$\xleftarrow{x\to 0} A+\underbrace{\frac{B}{x}}_{\pm\infty}+\frac{Cx}{x-1}=\underbrace{\frac{1}{x(x-1)}}_{\pm\infty}\to x\to 0$$

$\boxed{3}$ $$x^2\left(\frac{A}{x}+\frac{B}{x^2}+\frac{C}{x-1}\right)=\frac{1}{x^2(x-1)}x^2$$

$$xA+B+\frac{Cx^2}{x-1}=\frac{1}{x-1}$$

$$\frac{A}{2}-\frac{1}{4}+\frac{1}{2-1}=\frac{1}{4\cdot 1}$$

$A=-1$

$\int_{-\infty}^{b} f(x)\,dx \overset{def}{=} \lim_{a\to+\infty} \int_a^b f(x)\,dx$ Se lim ∃ finito

$f: (-\infty, b] \to \mathbb{R}$ integ su $[a\ b]$ $\forall\, a<b$

$\int_{-\infty}^{0} e^{-x^2}\cdot x\,dx$

$\int_a^0 e^{-x^2} x\,dx$ $x^2 = t$

$2x\,dx = dt$

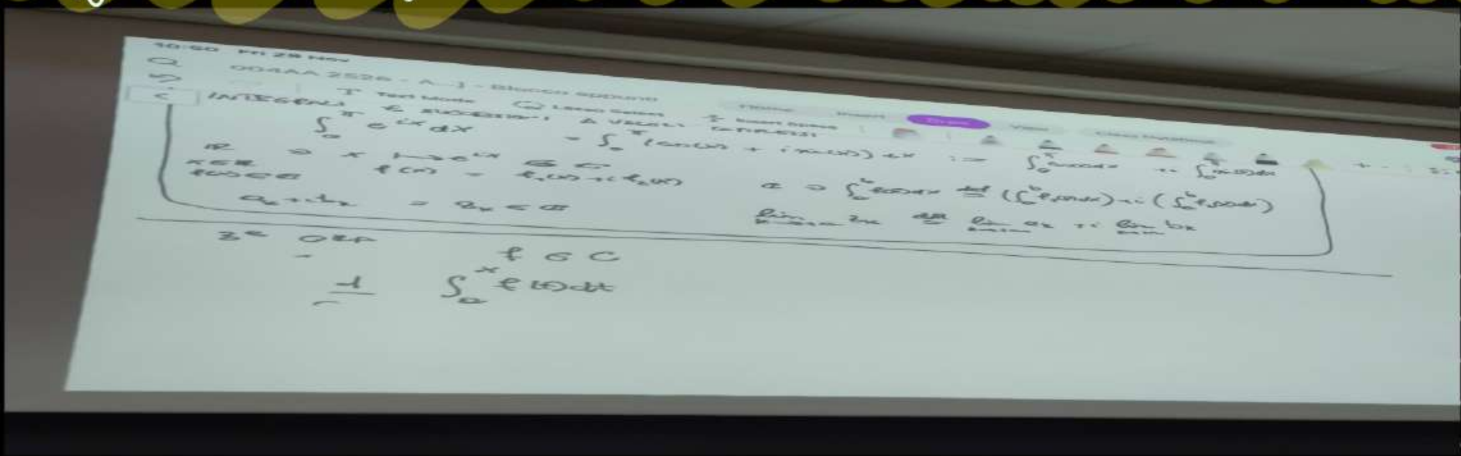
$\frac{1}{2}\int_a^0 e^{-t}\,dt = -\frac{1}{2}\int_0^{a^2} e^{-t}\,dt = +\frac{1}{2} e^{-t}\Big|_0^{a^2} = \frac{1}{2}(e^{-a^2}-1)$

$\lim_{a\to-\infty}\int_a^0 x e^{-x^2}\,dx = \lim_{a\to-\infty}\left(\frac{1}{2}(e^{-a^2}-1)\right) = -\frac{1}{2}$

INTEGRALI E SUCC A VALORI COMPLESSI

Si spezza int

$\int_0^\pi e^{ix}\,dx = \int_0^\pi [\cos(x) + i\,sen(x)]\,dx := \int_0^\pi \cos(x)\,dx + i\int_0^\pi \sin x\,dx$



3 ora

$\frac{d}{dx}\left(\int_a^x f(t)\,dt\right) = f(x)$ Teo fond calcolo

f cont deriv $G' = f$ $\int_a^b f(t)\,dt = G(b) - G(a)$

$\frac{d}{dx}\left[\int_a^{\beta(x)} f(t)\,dt\right.$ $\int_a^{\beta(x)} f(t)\,dt = G'(\beta(x)) - G(a)$

$\frac{d}{dx}\Big[$

Formula DERIV. per integrali con ESTREMI funzioni in x

$$\frac{d}{dx}\left(\int_{\alpha(x)}^{\beta(x)} f(t)\,dt\right) = f(\beta(x))\beta'(x) - f(\alpha(x))\alpha'(x)$$

$$\lim_{x\to 0^+} \frac{\int_0^{x^2} \frac{\sin(t)}{t}\,dt}{x} \qquad \left[\frac{0}{0}\right]$$

$f'(x) = \frac{\sin(x)}{x}$

$f(x) = \int_0^x \frac{\sin t}{t}\,dt$

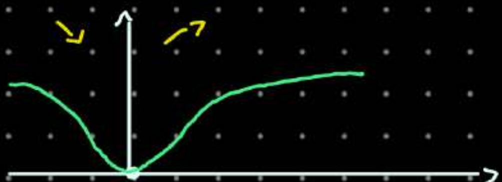
$$\text{hôp} \quad \frac{\frac{d}{dx}\left(\int_0^{x^2} \frac{\sin(t)}{t}\,dt\right)}{1} = \frac{\frac{\sin(x^2)}{2}\cdot 2x}{1} \xrightarrow{x\to 0} 0$$

Funz INTEGRALE

ES STUDIARE $f(x) = \int_0^{x^2} e^{-t^2}\,dt \quad x \geq 0$

$f(0) = 0$

$f'(x) = \underset{\vee\atop 0}{e^{-x^4}} \cdot 2x \qquad \underline{- - \overset{}{0} + +}$ (0)



$f''(x) = e^{-x^4}(-4x^3)\cdot 2x + e^{-x^4}\cdot 2 = e^{-x^4}(2-8x^4) =$

$= 2e^{-x^4}(1-4x^4)$

$$\lim_{x\to+\infty} f(x) = \int_0^{+\infty} e^{-t^2}\,dt = \frac{\sqrt{\pi}}{2}$$

ES tricky

$$\int_{\mathbb{R}} f(x)\,dx = \int_{-\infty}^{+\infty} f(x)\,dx := \int_{-\infty}^{c} f(x)\,dx + \int_c^{+\infty} f(x)\,dx$$

Scelgo c qualsiasi se $\int_{-\infty}^{c}$ e $\int_c^{+\infty}$ $\exists$ Finiti

$$\lim_{a\to-\infty}\int_a^c f(x)\,dx + \lim_{b\to+\infty}\int_c^b f(x)\,dx$$

$f(x) = x \qquad \int_{-\infty}^{+\infty} f(x) = dx \qquad \int_{-a}^{0} x\,dx + \int_0^a x\,dx = \frac{x^2}{2}\Big|_{-a}^{0} + \frac{x^2}{2}\Big|_0^a =$

$= 0 - \frac{(-a)^2}{2} + \frac{a^2}{2} - 0 = \frac{a^2}{2} - \frac{a^2}{2} = 0$

$\lim_{a\to+\infty}$

$$\lim_{a\to+\infty}\left(\int_{-a}^{0} x\,dx + \int_0^a x\,dx\right) = 0 \qquad \int_0^a x\,dx = \frac{a^2}{2} \xrightarrow{a\to+\infty} +\infty$$

Integ secondo valore Princip. Cauchy (V.P.)

TEOR CONFRONTO (PER INTEGRALI) $f \geq 0$ CONDIZIONE SUFFICENTE

$$\int_1^{+\infty} \frac{1+\sin(x)}{x^2}\,dx \quad \text{ESISTE O NO?}$$

$$\lim_{b\to+\infty}\int_1^b \frac{1+\sin x}{x^2}\,dx \quad \text{Esiste finito?}$$

$$0 \leq \frac{1+\operatorname{sen} x}{x^2}$$

$$\int_a^b f(x)\,dx$$

SE $g(x) \geq f(x) \Rightarrow \int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx$

$\int_a^{+\infty} g(x)\,dx$ ESISTE $\Rightarrow \int_a^{+\infty} f(x)\,dx$ ESISTE

Dim $\int_a^{x_1} f(x)\,dx \leq \int_a^{x_2} f(x)\,dx \qquad F(x) = \int_a^x f(t)\,dt$ F è cresc se $f \geq 0$

$$\lim_{x\to+\infty} F(x) = \begin{cases} +\infty & \text{se F non limit} \\ c & \text{se F limit} \end{cases}$$

$$0 \leq F(x) = \int_a^x f(t)\,dt \leq \int_a^x g(t)\,dt \leq \int_a^{+\infty} g(t)\,dt < +\infty$$

$f(x) = \frac{1+\sin(x)}{x^2} \geq 0$ se troviamo g : $g(x) \geq f(x)$

$\int_1^{+\infty} g(x)\,dx$ ESISTE

$$0 \leq \frac{1+\sin x}{x^2} \leq \frac{2}{x^2}$$

$$f(x) \leq g(x)$$

$\int_1^{+\infty} \frac{2}{x^2}\,dx < +\infty$ SIAMO A POSTO!

$$= \lim_{b\to+\infty}\int_1^b = 2\lim_{b\to+\infty}\int_1^b \frac{1}{x^2}\,dx$$

$\int_a^b f(x)\,dx$ $\quad\quad$ $\int_a^b c\cdot f(x)\,dx = c\int_a^b f(x)\,dx$

$$\int_a^b f(x)+g(x) = \int_a^b f(x) + \int_a^b g(x)$$

$$\int_a^{+\infty} c\, f(x)\,dx = \lim_{b\to+\infty}\left(c\int_a^b f(x)dx\right) = c\cdot \lim \text{---}$$

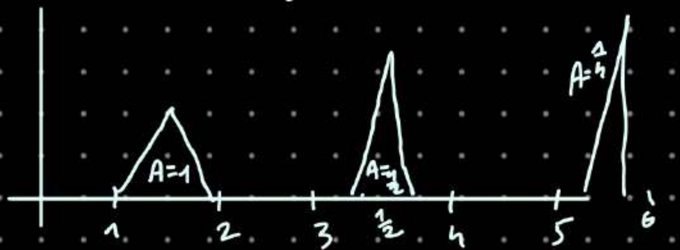
$$\int_a^{+\infty} (f+g)(x)\,dx = \int_a^{+\infty} f + \int_a g$$

$f \geq 0$ $\quad\quad$ $f(x) \geq g(x) \geq 0$

$$\lim_{b\to+\infty}\int_a^b g(x)\,dx = +\infty \implies \lim_{b\to+\infty}\int_a^b f(x)\,dx = +\infty$$

se non converge

→ Condiz. Suffic.



$$\int_0^{+\infty} f(x)\,dx = \sum_{k=0}^{\infty}\frac{1}{2^k} \quad\quad \sup f = +\infty$$

$f(x) \leq g(x)$

ES PER CASA → STUD conver.

- $\displaystyle\int_2^{+\infty}\frac{1+x+x^2}{1+x+x^3}\,dx$

- $\displaystyle\int_2^{+\infty} e^{-x^3}\,dx$